BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

STUDY PROJECT · PHASE 3

Explainable Machine Learning Framework for Battery Health Assessment and Dynamic Backup-Time Prediction

IMPLEMENTATION AND VALIDATION · REVISION 2 FINAL SUBMISSION

Verified package: 5,593 records · 3 real public datasets · 2,532 held-out predictions
22 graphs · 8 persisted models · 20 automated tests executed

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Evidence statement and declaration

What this report proves
The submitted software executes an end-to-end, leakage-aware workflow for capacity/SOH estimation, formula and machine-learning backup-time comparison, explainability, persistence, rule-based guidance, and dashboard reporting. Quantitative evidence is reported separately for the controlled CSGB benchmark and real public laboratory data from NASA PCoE, CALCE, and TRI/Matr.io.

What this report does not claim
The results are not field validation, deployment certification, a certified battery-management system, or safety authority. TRI expands real laboratory evidence across 94 aged cells and eight constant-current rates, but it is not dynamic field telemetry. The Oxford adapter is fixture-tested; no Oxford quantitative result is claimed because its official MAT file was not processed.

Student declaration

We declare that this Phase-3 submission presents the implemented project evidence available in the accompanying ZIP. Metrics, figures, models, aligned predictions, and test evidence are reproducible from the included code and processed inputs. Public source ownership and licensing remain with their respective providers; raw public archives are not redistributed.

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Abbreviations

Term	Meaning	Term	Meaning
Ah	Ampere-hour	MAE	Mean Absolute Error
SOH	State of Health	RMSE	Root Mean Squared Error
M4	Formula baseline	MAPE	Mean Absolute Percentage Error
M5	Capacity/SOH model	R^2	Coefficient of determination
M6	Backup-time ML model	SHAP	SHapley Additive exPlanations
LOBO	Leave-one-battery-out	CSGB	Controlled Synthetic Golden Benchmark

1. Executive summary

Phase 3 implements and validates the framework proposed in Phase 1 and designed in Phase 2. Revision 2 expands the real experimental evidence with the official TRI/Matr.io 94-cell multi-rate validation dataset. The final package contains a modular Python pipeline (M1–M10), prepared evaluation data, aligned record-level predictions, regression and classification metrics, global/local Shapley explanations, eight persisted models, twenty-two comparison figures, a Streamlit interface, and automated verification evidence.

Dataset	Evidence	Rows	Train / holdout	M5 MAE (Ah)	M6 MAE (min)	Group overlap
CSGB	controlled synthetic	720	540 / 180	0.056	1.519	0
NASA	real experimental	579	298 / 281	0.074	2.826	0
CALCE	real experimental	3,544	1,665 / 1,879	0.029	1.721	0
TRI	real experimental	750	558 / 192	0.122	34.638	0

- 5,593 validated cycle-level records were used: 720 CSGB controlled records and 4,873 real laboratory records from NASA, CALCE, and TRI.
- All four evaluations used battery-group holdouts with zero development/holdout battery overlap.
- The M5 capacity model achieved MAE 0.0560 Ah on CSGB, 0.0737 Ah on NASA, 0.0287 Ah on CALCE, and 0.1222 Ah on TRI.
- The M6 runtime model achieved MAE 1.519, 2.826, 1.721, and 34.638 minutes respectively; TRI's long low-rate tests make absolute-minute errors larger, so MAPE and rate-stratified results are also reported.
- The transparent M4 operational formula was stronger by aggregate MAE on NASA, CALCE, and TRI, while M6 was stronger on the varied-load CSGB benchmark. On TRI, M6 nevertheless achieved lower aggregate MAPE and lower MAE at five of eight rate levels. These mixed results are retained rather than selectively reported.
- Eight models reloaded within a 1×10^{-9}-minute numerical tolerance. Global and local Shapley outputs passed for every evaluated domain; the supplied execution used the documented deterministic fallback because the optional SHAP package was unavailable in the build runtime.
- The complete validation suite executed 20 automated tests: 19 passed and one Streamlit AppTest was explicitly skipped because Streamlit was unavailable in the build runtime. Dashboard backend, input guards, syntax, outputs, and the full analytical pipeline passed; the pinned requirements enable the AppTest on a fully installed environment.

Principal finding
A nonlinear model is not automatically superior. M6 reduced runtime MAE by 82.66% versus M4 on varied-load CSGB. M4 had lower aggregate MAE on NASA, CALCE, and TRI; however, TRI M6 had lower MAPE (3.34% versus 3.56%) and lower MAE at five of eight discharge rates. Protocol diversity and deployment-domain data matter more than model complexity alone.

2. Phase continuity, objectives and scope

2.1 Continuity from Phases 1 and 2

Phase	Role	Evidence carried forward
Phase 1	Problem and research definition	Title, integrated health/runtime objective, public-data commitment, explainability, formula benchmark, non-safety boundary.
Phase 2	Design and controlled proof of concept	M1–M10 contracts, common schema, leakage-safe workflows, CSGB benchmark, record alignment, comparison graphs and test design.
Phase 3	Implementation and validation	NASA/CALCE/TRI processing, grouped real-data evaluation, global/local Shapley attribution, persistence, dashboard, automated tests, and submission artefacts.

2.2 Implemented objectives

- Import source-adapted battery data and validate it against the common schema.
- Estimate remaining capacity and calculate SOH without using capacity-derived target leakage as an input.
- Calculate a transparent load-aware formula baseline and separately predict backup time using Random Forest regression.
- Compare M4 and M6 on identical held-out record IDs using MAE, RMSE, MAPE, bias and R^2.
- Provide built-in importance, permutation importance and SHAP global/local explanations.
- Persist models and verify reload/inference consistency.
- Expose validation, predictions, caveats, plots and downloads in Streamlit.
- Map model outputs to configurable, non-safety-critical guidance through M9.

2.3 Scope boundary

Included	Excluded / not claimed
Cycle-level lithium-ion laboratory data; capacity/SOH; discharge backup time; formula-versus-ML comparison; explainability; dashboard prototype; reproducible exports.	Cell balancing, SOC control, electrochemical diagnosis, safety interlocks, certification, production deployment, field-load validation, warranty decisions, or expert maintenance authority.

3. Implemented system architecture

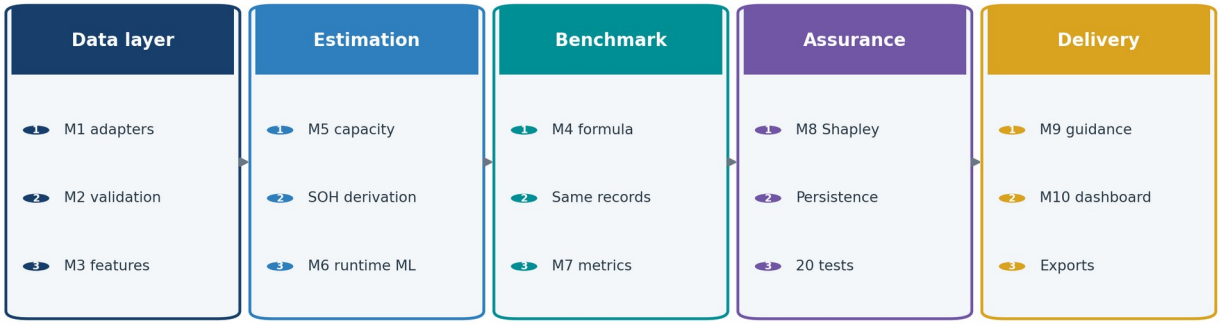


Figure 1. Implemented Phase-3 architecture and assurance flow.

3.1 Module responsibilities

Module	Implemented responsibility	Primary output
M1	Load common-schema inputs and source-adapted records.	Standardized cycle table
M2	Check schema, types, duplicates, nulls and physical ranges.	Clean table + quality report
M3	Create cycle, thermal, current, voltage, load and resistance features.	Leakage-safe feature matrices
M4	Calculate reference and operational formula runtime.	Formula runtime estimates
M5	Fit capacity Random Forest and derive SOH.	Predicted capacity + SOH
M6	Fit a separate dynamic backup-time Random Forest.	ML runtime estimate
M7	Align targets/predictions and calculate errors/residuals.	Metrics + comparisons
M8	Generate built-in, permutation and SHAP explanations.	Global/local attribution
M9	Apply configurable, secondary guidance rules.	Health class + recommendation
M10	Present validation, estimates, plots, caveats and downloads.	Streamlit interface + exports

3.2 Record-level traceability

Every held-out prediction export retains dataset, record_id, battery_id, cycle_count, actual targets, M4 formula outputs, M5 capacity/SOH outputs, M6 runtime output, residuals, M9 labels and recommendations. This is the implemented bridge from module output to each graph and aggregate metric.

4. Dataset provenance and evidence domains

4.1 Evidence classification

Dataset	Nature	Batteries	Usable rows	Phase-3 use	Status
CSGB v1.0	Controlled synthetic benchmark	12	720	Continuity, known-answer and varied-load software benchmark	Evaluated
NASA PCoE B0005/06/07/18	Real public laboratory	4	579	Capacity/SOH and runtime held-out validation	Evaluated
CALCE CS2_35/36/37/38	Real public laboratory	4	3,544	Capacity/SOH and runtime held-out validation	Evaluated
TRI ResVal multi-rate	Real public laboratory	94	750	Cross-cell capacity/SOH and rate-dependent runtime validation	Evaluated
Oxford Battery Degradation Dataset 1	Real public laboratory	8 in source	—	Native MAT adapter contract and fixture	Adapter only

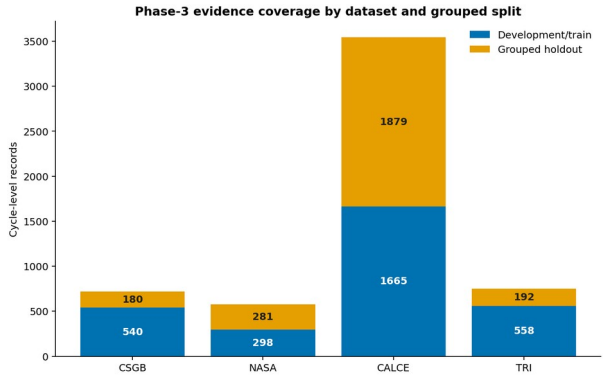


Figure 2. Record coverage by domain and split.

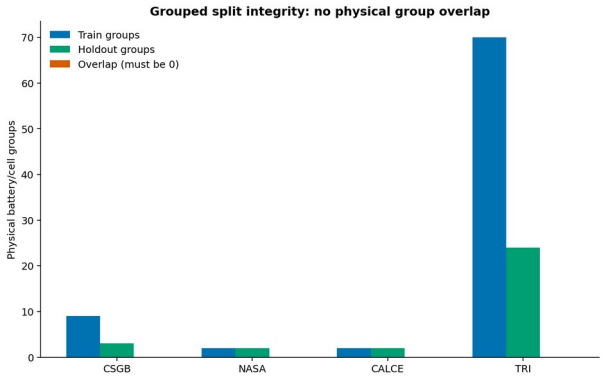


Figure 3. Battery-group split integrity.

4.2 Source processing and provenance

- NASA: official PCoE MAT files were processed into discharge-cycle summaries. Capacity and runtime use measured discharge data; the most recent preceding impedance observation supplies Re+Rct where available.
- CALCE: official CS2 Excel workbooks were processed into discharge-cycle summaries. Capacity is derived from the change in cumulative discharge capacity; runtime comes from elapsed discharge time; resistance is an explicitly labelled voltage-sag proxy.
- TRI: the official ResValData Maccor archive was streamed directly and converted to 750 rate-characterization discharge summaries from 94 aged 4.84 Ah Panasonic 21700 cells. Capacity, energy, representative load power, and time to the 2.7 V cutoff are derived from observed samples.
- CSGB and Oxford: the Phase-2 CSGB workbook is retained unchanged for controlled continuity checks. Oxford Cell*/cyc*/C1dc parsing is fixture-tested, but no Oxford metric is claimed because the official MAT file was not processed.

4.3 TRI extraction rules and limitations

The TRI ResVal subset contains eight constant-current characterization conditions: C/80, C/40, 0.05 A, C/10, C/7, C/5, 1C, and 2C. The adapter selects the first complete discharge to 2.7 V at each rate. Ninety-three cells supply all eight records; cell 0000ED supplies six low-rate records and no usable 1C/2C record, consistent with the authors' released processing code. No replacement records are fabricated.

The raw files do not contain measured temperature or prior aging-cycle count. The common-schema adapter therefore uses a declared constant 25 °C placeholder and a zero cycle-count sentinel; both are excluded from physical interpretation. TRI expands the number of real cells and the discharge-rate range, but it remains controlled laboratory evidence rather than field telemetry.

Raw-data distribution
The final ZIP includes compact processed cycle summaries, quality metadata, and provenance hashes. Large raw public archives are not redistributed. NASA, CALCE, and TRI preparation scripts support authorized local reproduction from official source files; TRI's 250,042,781-byte archive is represented by its SHA-256 digest and derived 750-row summary.

5. Data preparation and leakage-safe evaluation

5.1 Common schema

Category	Fields used
Identity/provenance	record_id, battery_id, source_dataset, cycle_count
Operating conditions	temperature_c, discharge_current_a, avg_voltage_v, min_voltage_v, load_power_w
Battery condition	internal_resistance_ohm, rated_capacity_ah
Targets	measured_capacity_ah, backup_time_min
Engineered	c_rate, thermal_stress, resistance_voltage_drop_v, voltage_sag_v
Derived outputs	predicted_capacity_ah, predicted_soh_percent, formula_runtime, ml_runtime, residuals, M9 labels
TRI protocol metadata	source_rate_index, source_rate_label, source_nominal_c_rate, source_file

5.2 Leakage controls

- M5 excludes measured_capacity_ah, backup_time_min and any target-derived SOH value from the feature matrix.
- M6 excludes actual backup time and uses M5 predicted SOH, not actual SOH, as an optional health input.
- M4 operational comparison uses M5-predicted capacity; a separate M4 reference calculation using measured capacity is reported only as a formula verification diagnostic.
- Feature contracts are exported and automatically tested.

5.3 Grouped holdouts

Dataset	Development batteries	Held-out batteries	Rows	Overlap
CSGB	NASA2_B01–03; NMC5_B01–03; OXF074_B01–03	NASA2_B04; NMC5_B04; OXF074_B04	540 / 180	0
NASA	B0005, B0006	B0007, B0018	298 / 281	0
CALCE	CS2_35, CS2_36	CS2_37, CS2_38	1,665 / 1,879	0
TRI	70 deterministic physical-cell IDs	24 deterministic physical-cell IDs	558 / 192	0

The split is battery-grouped rather than a random row split. Therefore, adjacent cycles from the same physical battery cannot appear on both sides of the evaluation. This makes the result materially more credible than record-level random splitting, although it remains dataset- and protocol-specific.

6. Modelling, formula baseline and evaluation metrics

6.1 Models

Workflow	Algorithm	Configuration	Output
M5	Random Forest regressor	250 trees; max_depth=14; min_samples_leaf=2; random_state=42	Capacity (Ah); SOH derived as capacity/rated capacity × 100
M6	Random Forest regressor	300 trees; max_depth=16; min_samples_leaf=2; random_state=42	Dynamic backup time (min)

6.2 Transparent backup-time baseline

Backup time (min) = [Predicted capacity (Ah) × Representative voltage (V) × Efficiency ÷ Load power (W)] × 60

CSGB uses its configured efficiency value. NASA, CALCE, and TRI discharge records represent direct cell-level laboratory energy delivery, so efficiency is set to 1.0. The M4 reference calculation is an internal consistency check when it uses measured capacity; the operational comparison uses M5-predicted capacity. Neither is described as deployment ground truth.

6.3 Metric interpretation

Metric	Meaning	Preferred direction	Important caution
MAE	Average absolute error in the target unit	Lower	Best direct measure of typical error
RMSE	Square-root of mean squared error	Lower	Penalizes large errors more strongly
MAPE	Average absolute percentage error	Lower	Unstable near zero actual values
Bias	Mean signed prediction error	Near zero	Positive means overprediction
R^2	Variance explained relative to a mean predictor	Higher	Can be negative; not an error unit

7. Quantitative validation results

7.1 M5 capacity and SOH

Dataset	Capacity MAE (Ah)	RMSE (Ah)	MAPE	R^2	SOH MAE
CSGB	0.0560	0.1051	3.29%	0.995	2.60 pp
NASA	0.0737	0.0841	4.67%	0.700	3.68 pp
CALCE	0.0287	0.0406	3.47%	0.928	2.61 pp
TRI	0.1222	0.2100	3.56%	0.770	2.53 pp

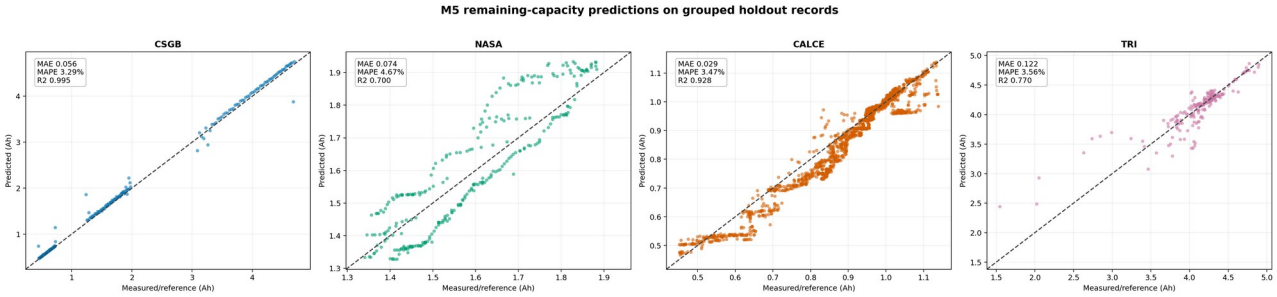


Figure 4. Actual versus predicted capacity on grouped holdouts.

CALCE produced the lowest absolute capacity error and the strongest real-data R^2. TRI extends the assessment from four batteries per source to 94 aged cells across eight rates; its 0.1222 Ah MAE corresponds to 3.56% MAPE and 2.53 percentage-point SOH MAE. CSGB's high R^2 reflects controlled construction and must not be treated as equivalent to real laboratory evidence.

7.2 SOH and runtime comparison

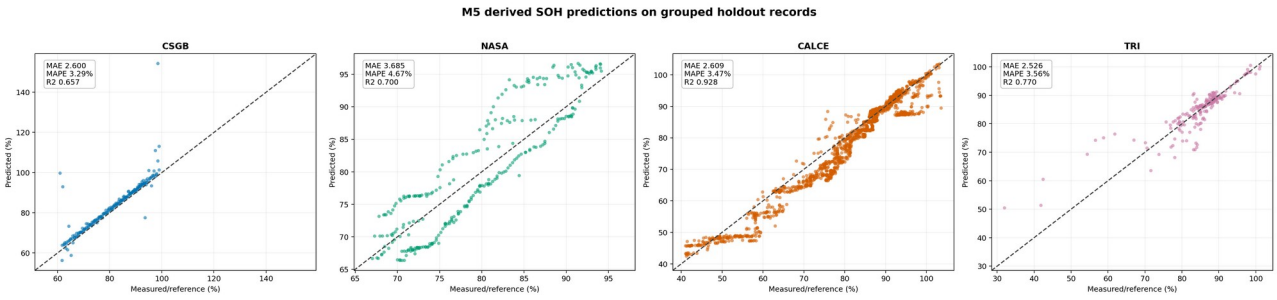


Figure 5. Actual versus predicted SOH on grouped holdouts.

Dataset	M4 MAE	M6 MAE	M4 MAPE	M6 MAPE	M6 MAE reduction	Lower MAE
CSGB	8.756	1.519	18.45%	2.91%	82.66%	M6
NASA	2.347	2.826	4.93%	5.97%	-20.38%	M4
CALCE	1.601	1.721	3.53%	3.76%	-7.51%	M4
TRI	33.681	34.638	3.56%	3.34%	-2.84%	M4

Interpretation
M6 is decisively better on the varied-load CSGB benchmark. NASA and CALCE use constrained protocols where a capacity-informed energy formula is strongly aligned with runtime. TRI adds a much wider C/80-to-2C rate range and long low-rate durations: M4 has slightly lower aggregate MAE, while M6 has lower MAPE and lower MAE at five of eight rate levels. Absolute-minute MAE is dominated by the longest tests, so aggregate and rate-level results must be read together.

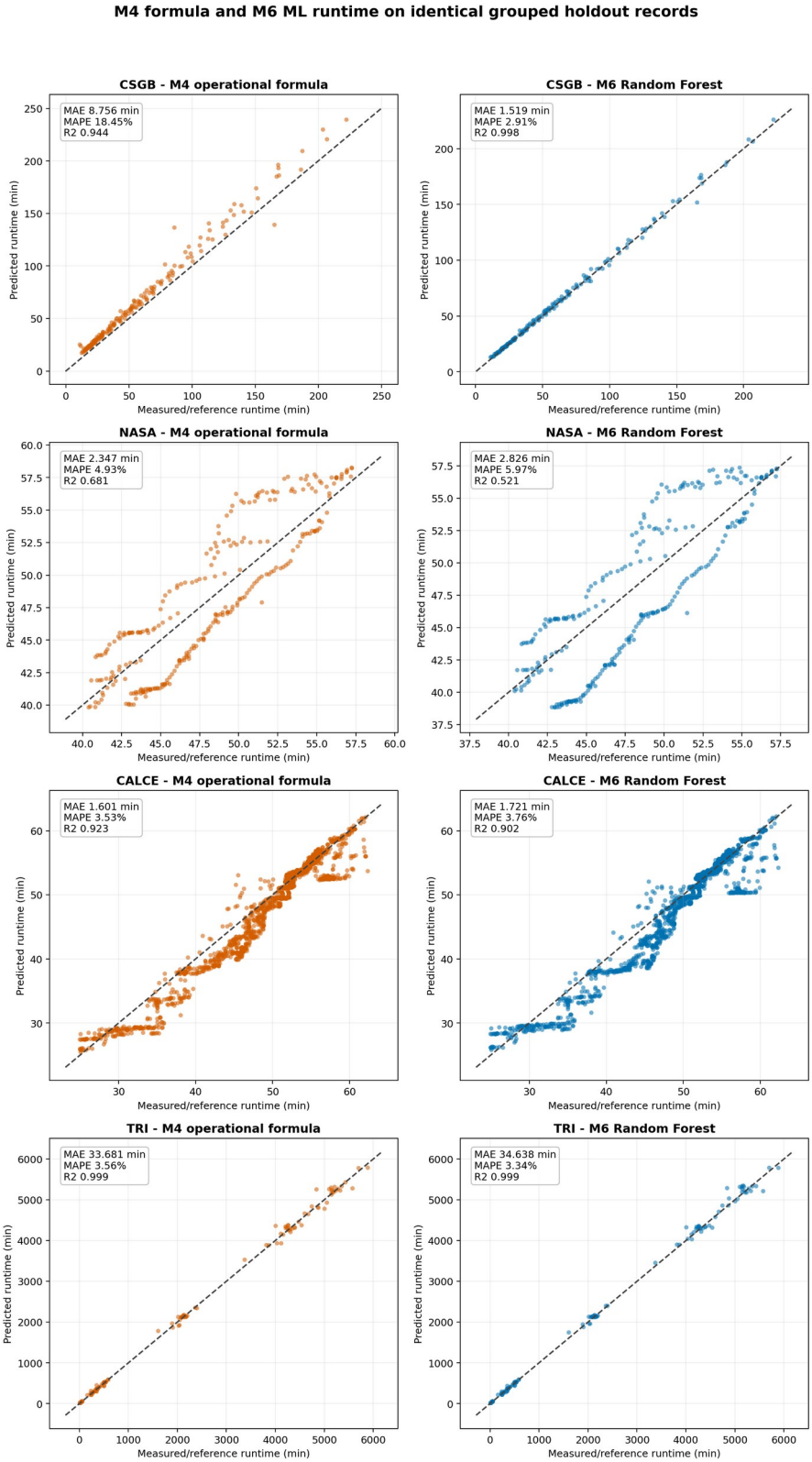


Figure 6. Formula and machine-learning runtime predictions on identical held-out records.

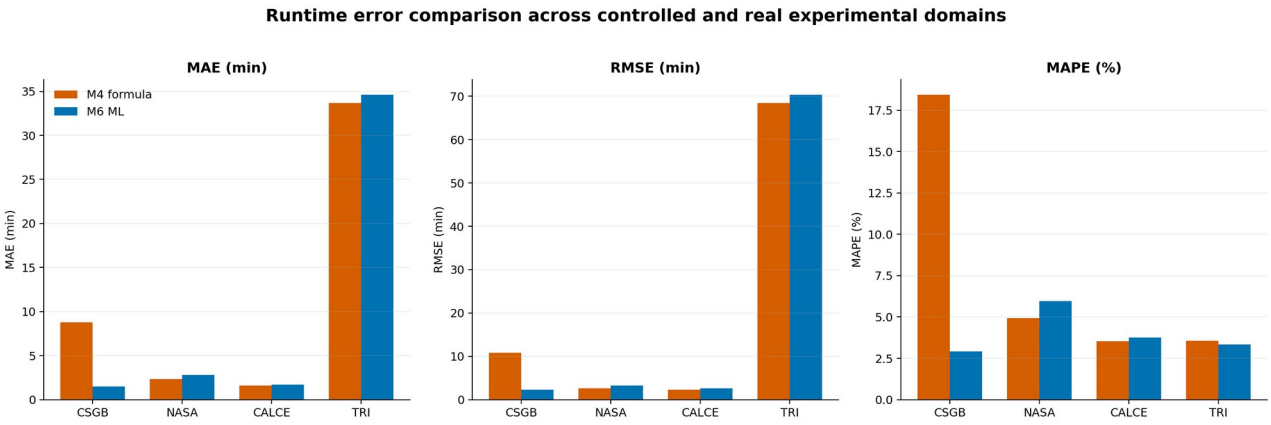


Figure 7. Runtime error metrics for M4 and M6.

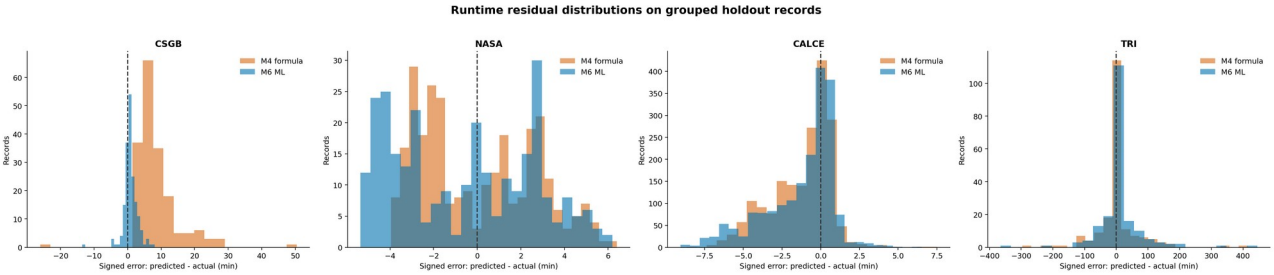


Figure 8. Runtime residual distributions.

7.3 Error stability across ageing

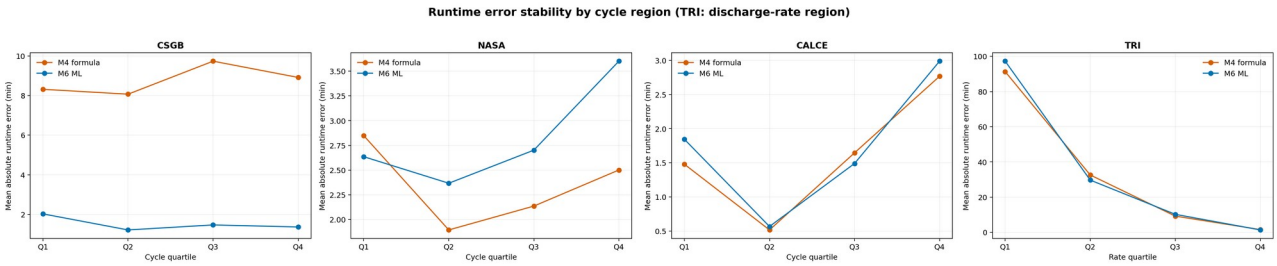


Figure 9. Runtime absolute error by cycle-age quartile.

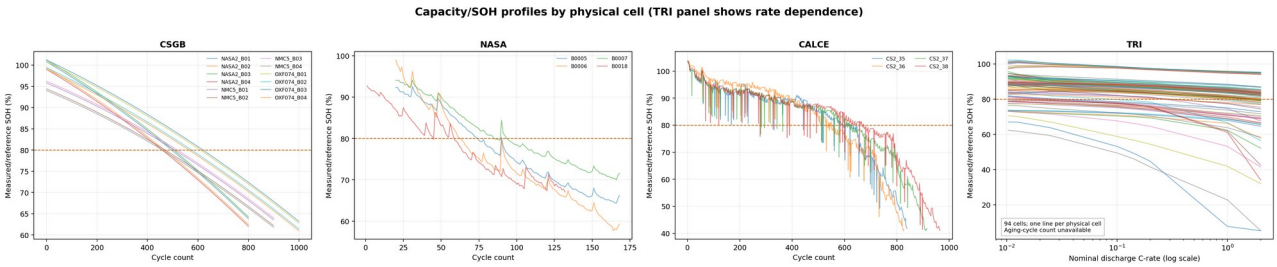


Figure 10. Held-out capacity degradation profiles.

The quartile and profile figures complement aggregate scores. For CSGB, NASA, and CALCE they show error across cycle-age bands and battery-specific degradation trajectories. TRI has no prior aging-cycle count in the raw files, so its panels use discharge-rate bands and rate-dependent capacity/SOH profiles instead of inventing an age sequence. Per-battery and per-rate metrics are exported for audit.

7.4 TRI rate-level comparison

Each TRI rate uses the same 24 held-out physical cells. The table shows why a single aggregate MAE is insufficient for this dataset.

Rate	Capacity MAE (Ah)	M4 MAE (min)	M6 MAE (min)	Lower MAE
C/80	0.090	89.978	81.535	M6
C/40	0.099	48.866	44.400	M6
0.05 A	0.077	92.711	113.000	M4
C/10	0.132	16.338	14.982	M6
C/7	0.130	11.254	12.017	M4
C/5	0.116	7.203	8.447	M4
1C	0.165	2.047	1.766	M6
2C	0.169	1.048	0.954	M6

M6 has lower MAE at C/80, C/40, C/10, 1C, and 2C; M4 is lower at 0.05 A, C/7, and C/5. The complete eight-row export is outputs/tri/rate_level_metrics.csv.

8. Explainability and model interpretation

M8 produces three complementary explanation layers: Random Forest built-in importance, held-out permutation importance, and global/local Shapley attribution. When the optional shap package is installed, the pipeline uses TreeSHAP. In the supplied build runtime that package was unavailable, so a deterministic Monte-Carlo Shapley fallback (24 permutations) generated the included global/local outputs under the same export contract. Built-in importance describes tree usage, permutation importance tests predictive dependence, and Shapley values attribute predictions relative to an expected value.

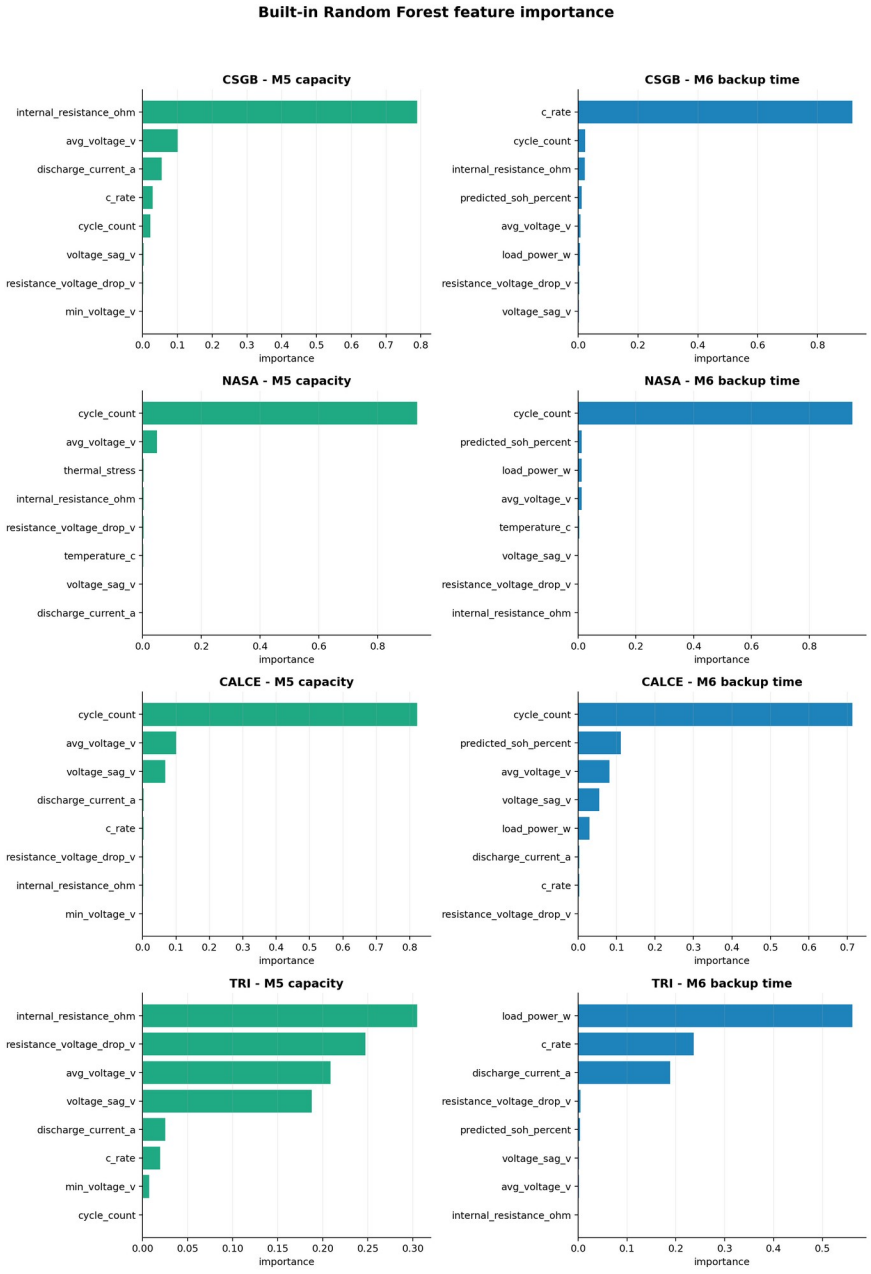


Figure 11. Built-in feature importance for M5 and M6.

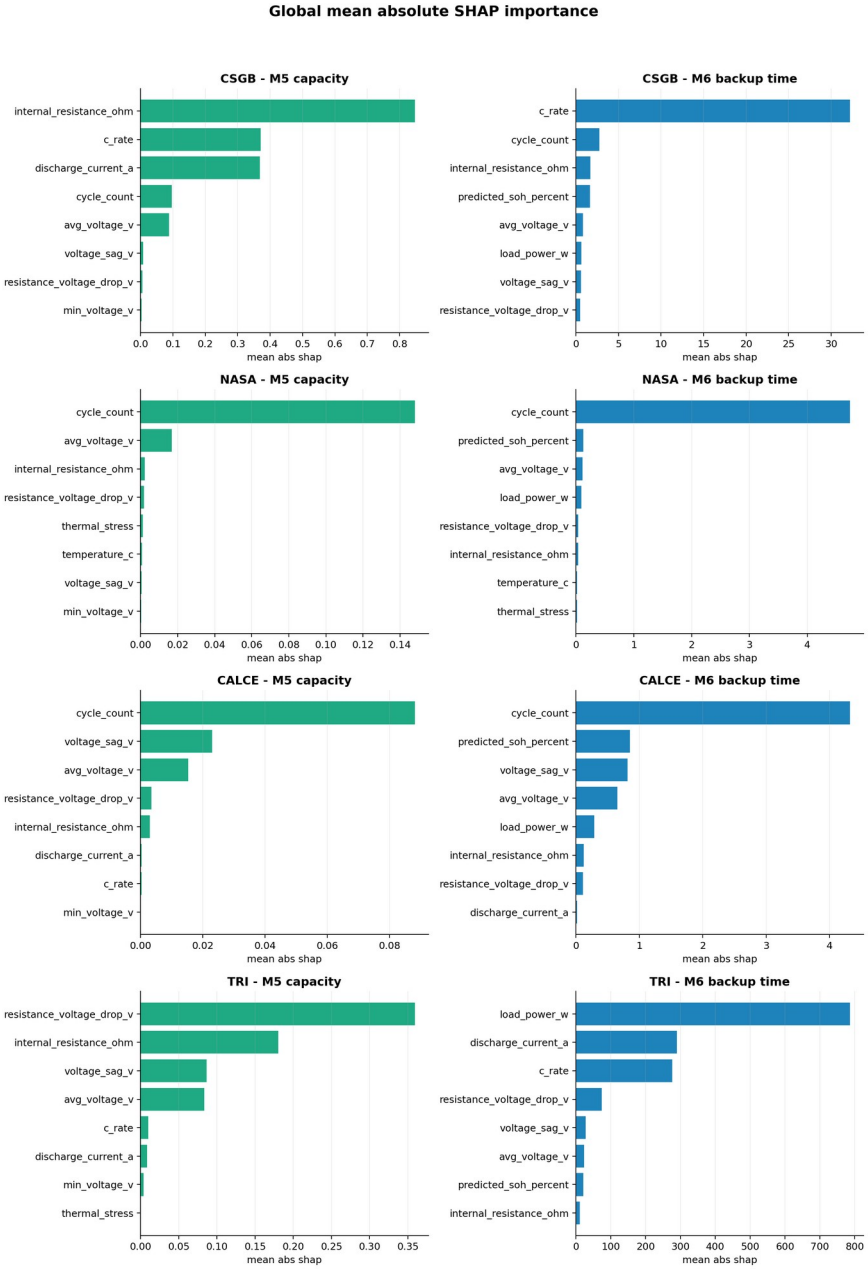


Figure 12. Mean absolute SHAP importance by evidence domain.

8.1 Global and local findings

Dataset	M5 dominant built-in feature	M6 dominant built-in feature	SHAP status	Rows explained
CSGB	Internal resistance (0.789)	C-rate (0.916)	Pass	180
NASA	Cycle count (0.934)	Cycle count (0.950)	Pass	200
CALCE	Cycle count (0.822)	Cycle count (0.713)	Pass	200
TRI	Internal resistance (0.305)	Load power (0.561)	Pass	192

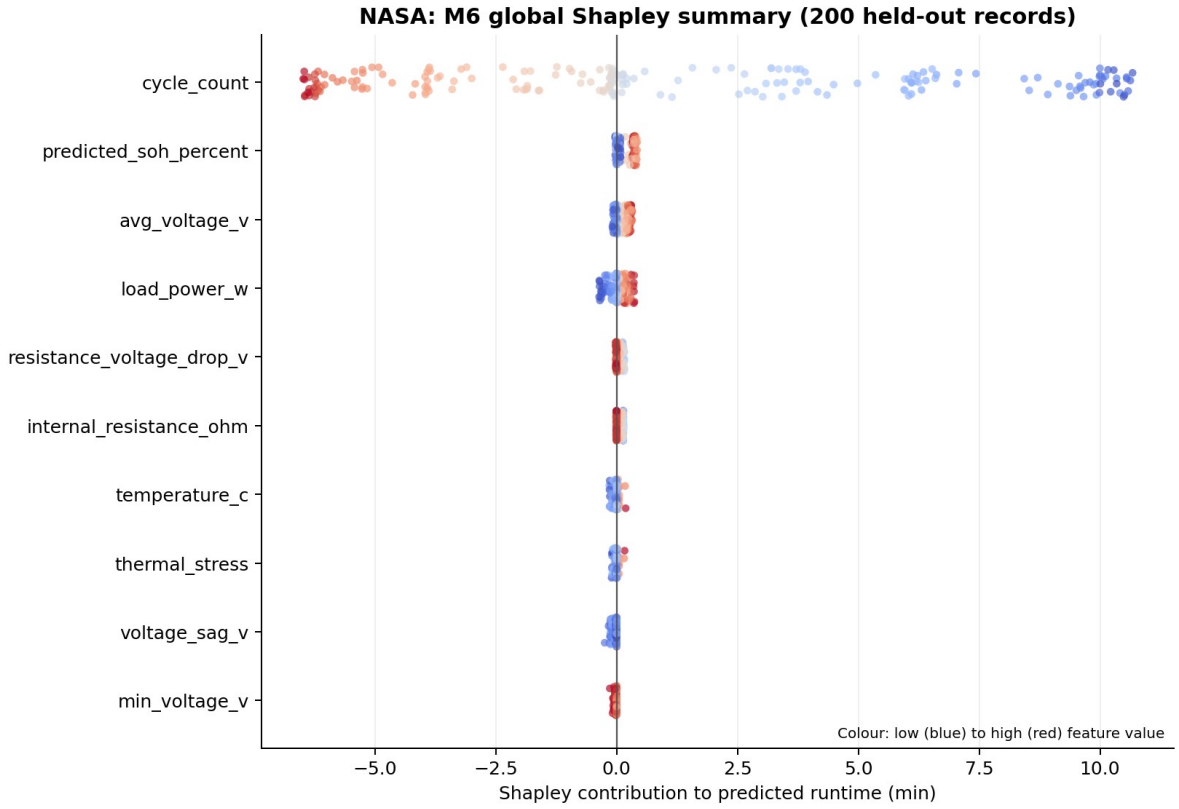


Figure 13. NASA M6 global SHAP summary.

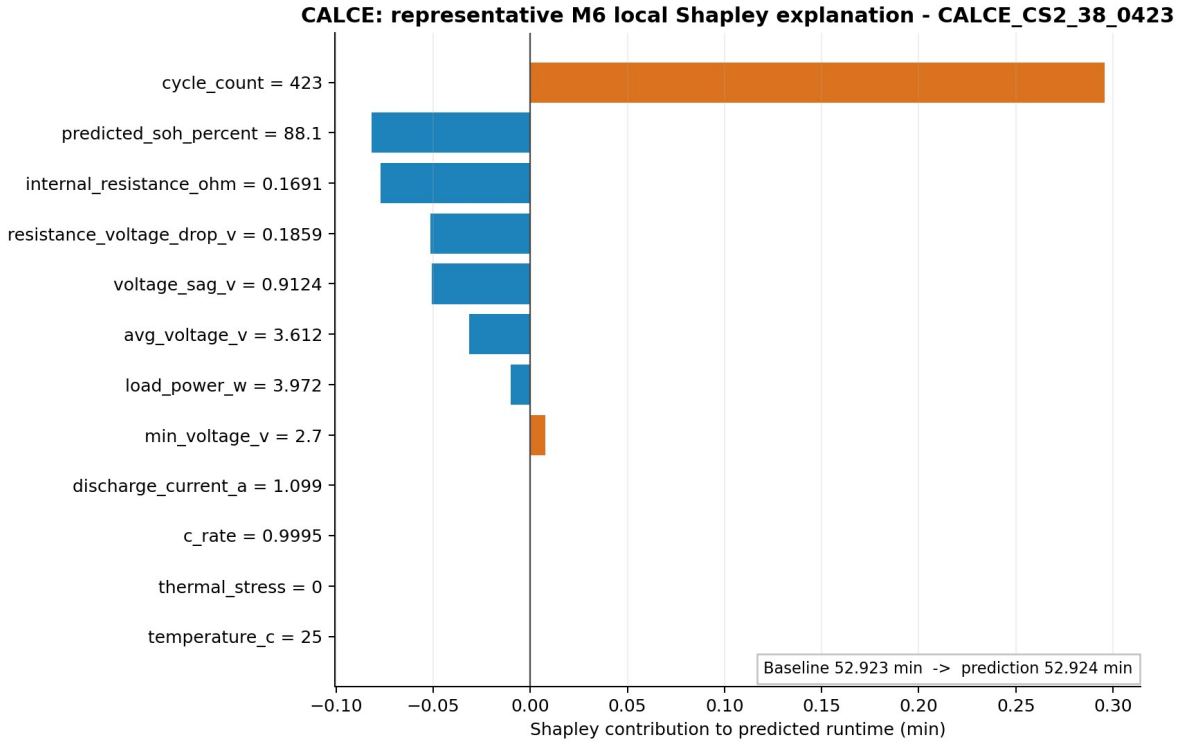


Figure 14. CALCE M6 local SHAP explanation.

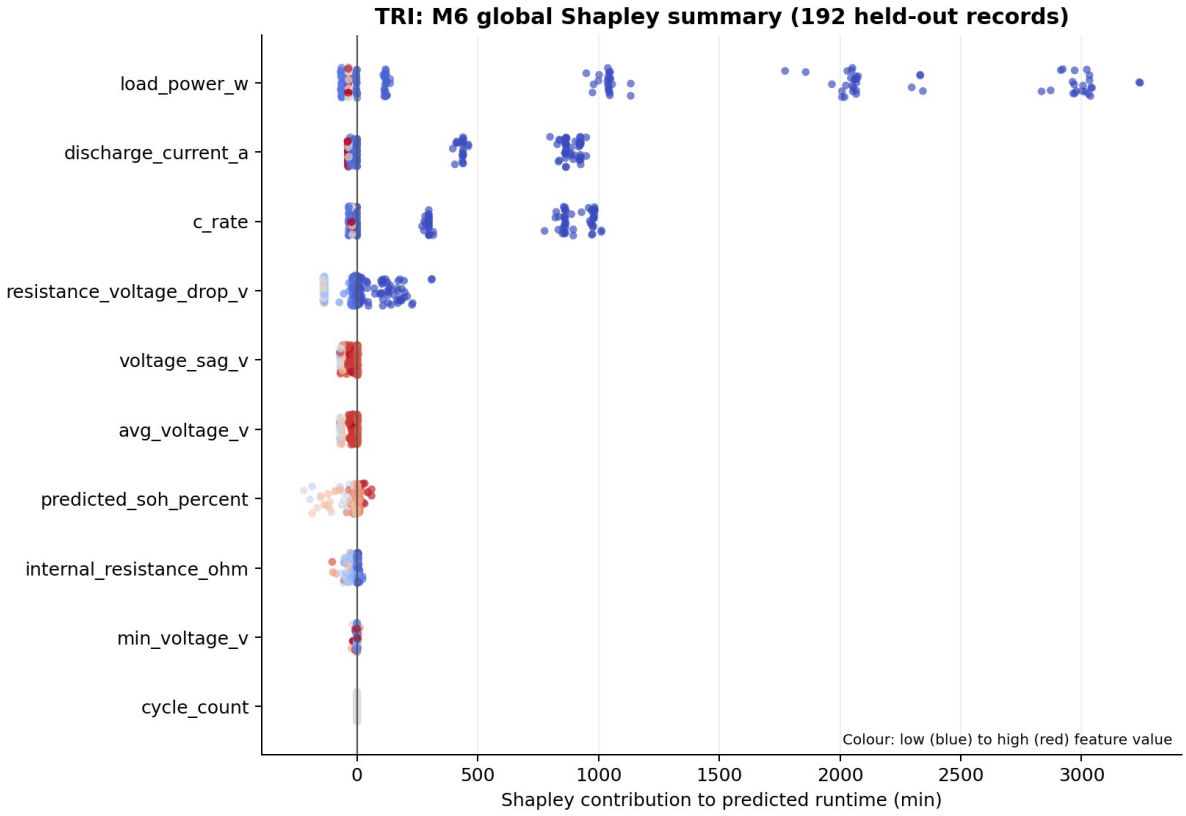


Figure 15. TRI M6 global Shapley summary.

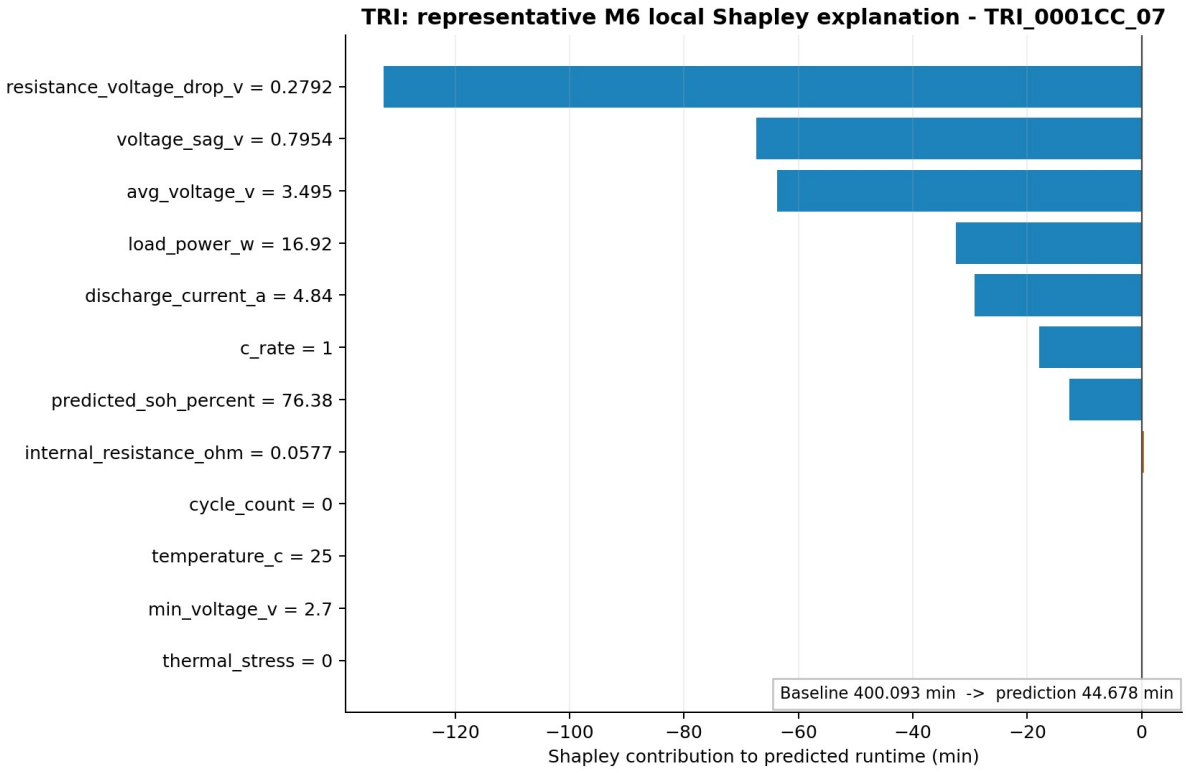


Figure 16. TRI M6 local Shapley explanation.

Feature dominance is domain-dependent. Resistance and C-rate dominate the varied CSGB construction; cycle count dominates the constrained NASA and CALCE protocols; and TRI uses resistance-related features for capacity while load power and C-rate dominate runtime. TRI's constant temperature and sentinel cycle count correctly receive no physical interpretation. Explanations must therefore be read within each source protocol.

9. Decision support and Streamlit dashboard

9.1 M9 decision support

M9 maps predicted SOH, available backup time, required runtime and data-quality flags to configurable labels and guidance. For conservative new-condition guidance, the dashboard uses the smaller of formula and ML runtime estimates. M9 remains secondary, non-safety-critical logic.

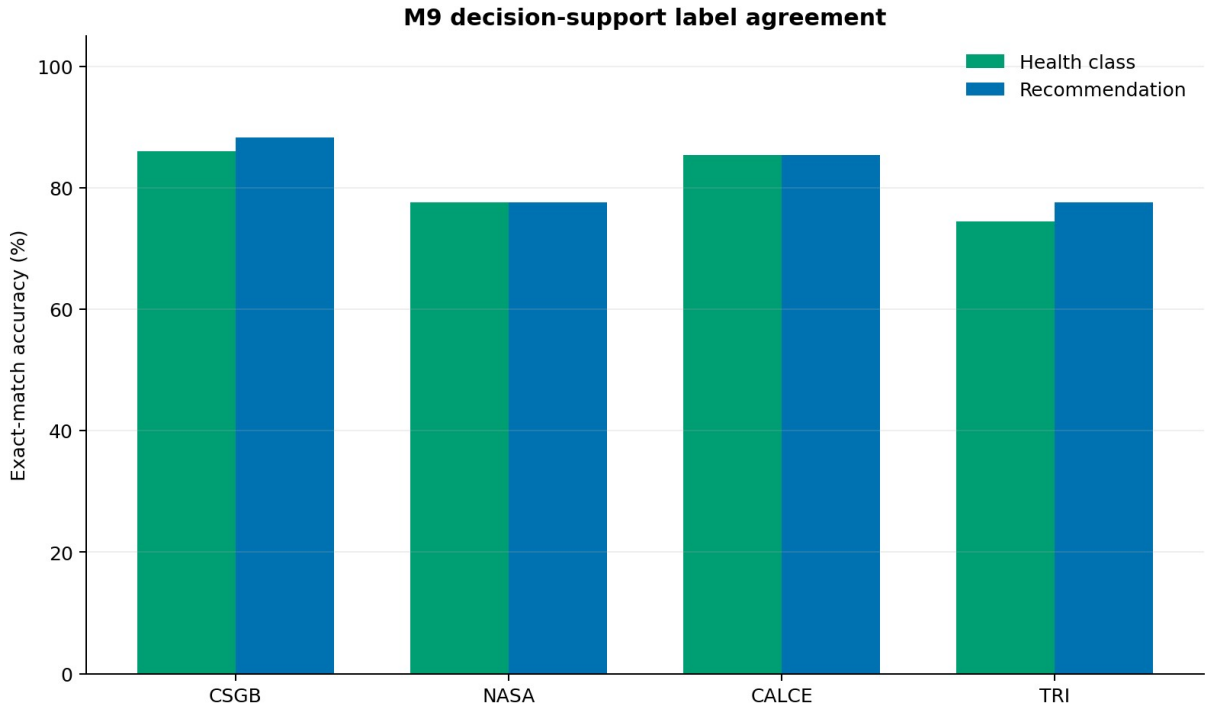


Figure 17. M9 exact-match agreement by dataset.

Reference-label caution
CSGB includes benchmark labels. NASA, CALCE, and TRI references are deterministic threshold labels derived from measured targets, not independent expert maintenance annotations. Their agreement scores validate rule wiring and sensitivity, not safety-classification accuracy.

9.2 M10 dashboard functions

Tab	Implemented function	Validation evidence
Overview	Evidence boundary, dataset inventory and key metrics	Backend/data checks + exported evidence
Data Quality	Quality, grouped-split and source-adapter status	Quality/split JSON + automated tests
Health and Runtime	Dataset results and comparison graphs	Backend/data checks + 22 PNG files
Explainability	Importance and global/local SHAP outputs	CSV/PNG contract + explanation tests
New Condition	Validated inputs; capacity, SOH, M4/M6 runtime and guidance	Backend smoke + guard tests
Downloads	Predictions, metrics and prepared data	Inventory and file-existence checks

10. Verification, persistence and performance

10.1 Automated validation

Layer	Observed result
Syntax compilation	Pass for implementation and tests
Standalone shared modules	M1–M9 regression checks pass
End-to-end pipeline	Pass on CSGB, NASA, CALCE, and TRI
Output contract	4 dataset outputs, 2,532 held-out rows, 22 graphs, 8 models
Automated unit/integration suite	20 executed: 19 pass, 1 dependency-aware skip
Streamlit application harness	AppTest skipped because Streamlit was absent; backend/guards/syntax pass

10.2 Model persistence

Dataset	Capacity max |Δ|	Runtime max |Δ|	Tolerance	Status
CSGB	1.78e-15	8.53e-14	1.00e-9	Pass
NASA	8.88e-16	2.84e-14	1.00e-9	Pass
CALCE	8.88e-16	2.84e-14	1.00e-9	Pass
TRI	2.66e-15	1.82e-12	1.00e-9	Pass

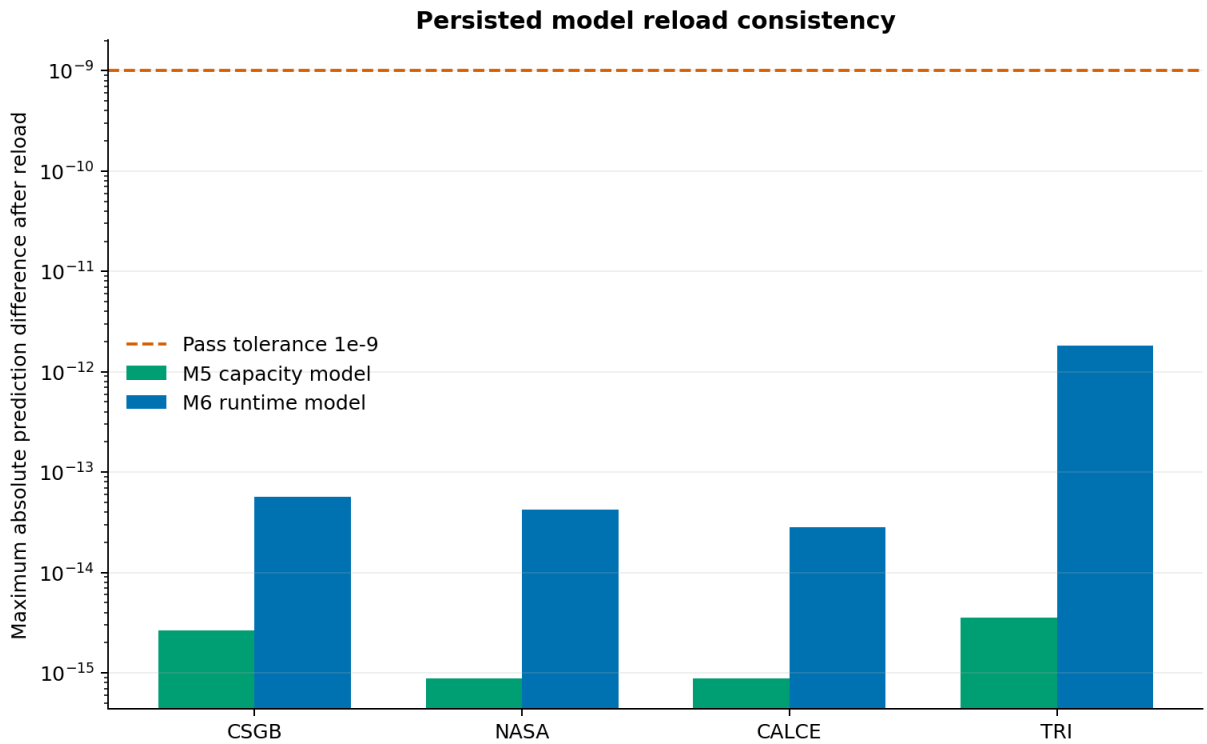


Figure 18. Persisted-model reload consistency.

10.3 Measured inference performance

Dataset	M5 inference (ms/row)	M6 inference (ms/row)
CSGB	0.944	0.492
NASA	0.748	0.356
CALCE	0.118	0.047
TRI	1.128	0.800

Times are observations from the validated environment, not service-level guarantees. The expanded four-domain pipeline completed in 87.25 seconds in the supplied execution environment.

10.4 Automated test coverage

Tests 1–10	Tests 11–20
CSGB continuity dataset	Model persistence/reload
NASA real dataset	Global/local Shapley outputs
CALCE real dataset	Graph/model inventory
TRI 94-cell dataset	TRI rate-level output
Zero group overlap	Dashboard backend smoke
Oxford native-contract fixture	Dashboard input guards
Target-leakage exclusion	Power-consistency warning
Formula known answer	Conditional Streamlit AppTest
Separated metric domains	No overclaim on real runtime result
Prediction export contract	Truthful adapter status

11. Discussion, limitations and residual risks

11.1 What the evidence supports

- The modular workflow is executable, traceable and reproducible from prepared inputs.
- Battery-group holdouts prevent same-battery cycle leakage in the reported evaluations.
- Capacity/SOH prediction is effective on the selected NASA, CALCE, and TRI laboratory cells, with source- and protocol-specific variation.
- Formula and ML runtime estimates can be fairly compared on the same records, and the winning method depends on the load protocol.
- Global and local explanations, persistence, dashboard behavior and output contracts are verified.

11.2 Limitations

Limitation	Effect	Mitigation / next evidence
Few physical batteries per real dataset	Uncertainty around population-level generalization	Add more cells, LOBO folds and chemistry/protocol strata
NASA/CALCE/TRI discharge protocols are constrained	Limited evidence for highly dynamic inverter loads	Evaluate dynamic-load profiles and field-like power traces
CALCE temperature represented as 25°C metadata	Cannot infer true thermal response	Use time-resolved temperature datasets
CALCE resistance is a voltage-sag proxy	Not directly comparable with EIS resistance	Keep provenance flag; add source-specific features
Oxford official file not processed	No Oxford metrics or graphs	Run included adapter on authorized source file and report separately
M9 real-data labels are deterministic	Not expert-grounded recommendation validation	Obtain domain-expert labels and calibrate thresholds
No field or safety validation	Prototype cannot authorize operational decisions	Controlled field trial, uncertainty calibration and safety review
TRI lacks temperature and prior aging-cycle count	No TRI thermal or age-trajectory conclusion	Retain provenance flags; add characterization metadata or another compatible source

11.3 Residual-risk position

Submission boundary

The package is ready as an academic implementation-and-validation submission. It should be presented as a decision-support prototype whose strongest empirical evidence comes from grouped held-out public laboratory batteries—not as a finished commercial battery-management or safety system.

12. Deliverables, execution procedure and output inventory

12.1 Included deliverables

Deliverable	Quantity / location	Purpose
Final report	DOCX + PDF	Submission narrative and verified interpretation
Implementation	M1–M10 Python modules	Executable analytical and dashboard workflow
Prepared data	CSGB workbook + NASA/CALCE/TRI CSVs + provenance	Offline reproducibility without redistributing raw public archives
Aligned predictions	4 CSV files; 2,532 rows	Record-level audit and further analysis
Metrics/explanations	CSV + JSON outputs	Quantitative comparison and interpretation
Graphs	22 PNG files	Coverage, accuracy, residual, rate, SHAP, persistence, and M9 evidence
Persisted models	8 Joblib files	Reloadable M5/M6 models for four domains
Verification	20 tests + log + JSON + output inventory	Submission integrity

12.2 Windows execution

```
cd Phase3_Implementation
py -3.12 -m venv .venv
.venv\Scripts\python -m pip install -r requirements.txt
run_full_validation.bat
run_phase3_dashboard.bat
```

run_full_validation.bat regenerates predictions, metrics, models and graphs for all four domains, checks the output contract, and executes the automated suite. run_phase3_dashboard.bat starts the Streamlit interface after the pinned requirements are installed. Existing verified outputs are included, so the report can be reviewed without rerunning the models.

12.3 Expected successful output

- PHASE-3 VALIDATION COMPLETED
- PHASE-3 OUTPUT VERIFICATION: PASS
- Ran 20 tests … OK (skipped=1 when Streamlit is unavailable)
- VALIDATION SUITE STATUS: PASS
- Graphs: 22; models: 8; held-out prediction rows: 2,532

Submission check

Use the included MANIFEST_SHA256.csv and VALIDATION_SUMMARY.md to verify the ZIP. No manual insertion of graphs or replacement of placeholder results is required.

13. Conclusion and future work

Phase 3 closes the principal implementation items carried forward from Phase 2. Revision 2 adds a working TRI/Matr.io source path and 750 real multi-rate records from 94 aged cells to the existing NASA and CALCE evidence. The project now includes leakage-safe capacity/SOH and runtime workflows, a transparent operational formula baseline, aligned comparison exports, global/local Shapley explanations, model persistence, rule-based guidance, a Streamlit dashboard, and a passed analytical verification suite.

The most important result is methodological: model complexity did not guarantee lower runtime error. M6 learned the varied-load CSGB benchmark well; M4 retained lower aggregate MAE on the constrained NASA, CALCE, and TRI protocols; and TRI's rate-level analysis showed M6 winning five of eight rates and producing lower aggregate MAPE. This supports retaining both an auditable formula comparator and a learned model, with interpretation conditioned on protocol and error scale.

13.1 Recommended next work

1. Run the Oxford source-native adapter on the authorized official MAT file and publish Oxford-only results without merging incompatible feature definitions.
2. Add more physical batteries and repeat leave-one-battery-out or grouped cross-validation with fold-level confidence intervals.
3. Collect or identify genuinely dynamic-load discharge profiles representative of inverter and UPS use.
4. Calibrate predictive intervals and out-of-distribution warnings before any operational pilot.
5. Obtain expert review of M9 thresholds and preserve the non-safety boundary unless formal assurance work is completed.
6. Package the dashboard in a controlled deployment environment with versioned model cards and audit logging.

Final Phase-3 outcome

The capstone target—an integrated, explainable framework for battery-health assessment and dynamic backup-time prediction—has been implemented and validated as an academic prototype, with evidence and limitations presented transparently.

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Appendix A – Metric definitions and comparison mapping

Project output	Compared with	Purpose
M4 formula operational runtime	actual_backup_time_min	Transparent baseline accuracy
M5 predicted_capacity_ah	actual_capacity_ah	Remaining-capacity accuracy
M5 predicted_soh_percent	actual_soh_percent	Health-assessment accuracy
M6 ml_backup_time_min	actual_backup_time_min	Dynamic runtime accuracy
M9 predicted health class	health_class reference	Exact-match rule/label comparison
M9 recommendation	expected_recommendation reference	Exact-match guidance comparison

Metric formulas

Metric	Definition
MAE	mean(|prediction − actual|)
RMSE	sqrt(mean((prediction − actual)²))
MAPE	mean(|prediction − actual| / |actual|) × 100
Bias	mean(prediction − actual)
R^2	1 − Σ(actual − prediction)² / Σ(actual − mean(actual))²
Exact-match accuracy	matching labels / total labels × 100

Units

Capacity errors are in Ah; runtime errors are in minutes; SOH MAE is reported in percentage points (pp); MAPE and exact-match accuracy are percentages.

Appendix B – Graph and file inventory

No.	Graph	Analysis supported
01	Dataset record coverage	Evidence volume and split
02	Grouped split integrity	Train/holdout groups and overlap
03	Capacity actual vs predicted	M5 accuracy
04	SOH actual vs predicted	M5 health accuracy
05	Runtime actual vs predicted	M4/M6 record-level fit
06	Runtime error metrics	MAE/RMSE/MAPE comparison
07	Runtime residual distributions	Bias/spread
08	Runtime error by cycle quartile	Ageing stability
09	Capacity degradation profiles	Trajectory tracking
10	Built-in feature importance	Model structure
11	Permutation feature importance	Held-out dependence
12	SHAP mean absolute importance	Global attribution
13–16	Runtime Shapley summaries	Domain-specific global explanations
17–20	Runtime Shapley local plots	Representative record explanations
21	Persistence reload consistency	Serialization integrity
22	M9 exact-match accuracy	Guidance wiring

Key machine-readable files

File	Use
outputs/phase3_metrics.json	Complete dataset, split, model, explanation, persistence and timing metrics
outputs/phase3_metrics_table.csv	Compact cross-domain metric comparison
outputs/<dataset>/heldout_predictions.csv	Aligned record-level actual/predicted outputs
outputs/<dataset>/*feature_importance.csv	Built-in, permutation and SHAP global values
outputs/models/*.joblib	Eight persisted M5/M6 models
outputs/output_inventory.csv	Verified graph/model/data inventory
outputs/phase3_test_report.json	Machine-readable validation-suite status
TEST_EXECUTION_LOG.txt	Human-readable commands and observed results
outputs/tri/rate_level_metrics.csv	TRI held-out results separated across eight characterization rates